PracticePro

# Zero-Failure Launch Audit

VEGA (Legal) + Atrium (Property) | Codebase vs. Marketing Claims Reconciliation

**Date:** 14 May 2026

**Scope:** Full codebase audit across 4 dimensions

**Codebase:** React + Convex full-stack application


**Audit Summary**

This audit compared every marketing claim on the PracticePro landing page against the actual Convex backend functions, App.tsx routes, and component implementations. The findings reveal **4 Ghost Features** (marketed but non-existent), **13 Partial Features** (UI exists but key functionality missing), **7 Critical security vulnerabilities**, and **3 hardcoded API keys** exposed in the client bundle. Additionally, **rent payments are invisible in financial reports** due to a dual-system data inconsistency, and the core CRUD mutations have **zero authentication**.

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# 1. Feature Traceability Matrix

The following matrix compares every feature claim on the PracticePro landing page against actual code implementation. Each feature is traced through three layers: the App.tsx route, Convex backend functions, and React component code. Features are classified as LIVE (fully functional), PARTIAL (UI exists but incomplete), GHOST (marketed but no code), or WIRING_ONLY (backend exists but no user-facing UI). This analysis covers both VEGA (legal) and ATRIUM (property) product modes.

## 1.1 VEGA (Legal) Features

| Feature | Route | Convex Functions | Components | Status |
|---|---|---|---|---|
| Case Management | matters, matterDetail | getFirmData, getMatterDetails | MatterList (Kanban), MatterDetailView | LIVE |
| ALOA AI Copilot | Floating panel | generateContent, saveAloaMessage | AloaPanel, AloaChat | PARTIAL |
| Procedural Intelligence | research, indexer | None for judgments DB | MatterIntakeWizard, proceduralRules.ts | PARTIAL |
| Enterprise Jurisdiction Intake | matterIngestion modal | None specific | MatterIntakeWizard, SmartMatterModal | PARTIAL |
| Legacy Intelligence Activation | NONE | NONE | Landing page BentoCard only | GHOST |

Table 1: VEGA Feature Traceability

## 1.2 ATRIUM (Property) Features

| Feature | Route | Convex Functions | Components | Status |
|---|---|---|---|---|
| Portfolio Dashboard | dashboard, atriumEngine | getCashFlowSummary | Dashboard, RevenueEngine | LIVE |
| Automated Rent Collection | atriumEngine | addLedgerEntry, getCashFlowSummary | LedgerManager, CollectRentModal | PARTIAL |
| Lease Management | properties, propertyDetail | getPropertyDetails | PropertyManagerView, PropertyDetailView | PARTIAL |
| Maintenance Tracking | propertyDetail (tab) | Stored inline on property | PropertyTrackingView | LIVE |
| Secure & NDPA Compliant | Cross-cutting | verifyLogin, requireFirmUser | SecuritySettings, FeatureGuard | PARTIAL |

Table 2: ATRIUM Feature Traceability

## 1.3 Pricing Tier Feature Claims

| Feature | Tier Route | Convex Functions | Components | Status |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Revenue Ledger | atriumEngine | addLedgerEntry, getLedgerByFirm | LedgerManager | LIVE |
| WhatsApp Reminders | atriumEngine | sendWhatsApp, incrementWhatsAppQuota | AutomationCenter, ComposeModal | PARTIAL |
| Service Charge Tracking | atriumEngine | upsertServiceCharge, getDefaulters | ServiceChargeMonitor | LIVE |
| Legal Document Generation | propertyDetail | generateRentDemand | Integrated into property detail | PARTIAL |
| Tenant Vetting System | atriumEngine pipeline | addLeadToPipeline | VacancyPipeline (manual score only) | GHOST |
| WhatsApp Automation Engine | atriumEngine | runDailyAutomation (simulated) | AutomationCenter | PARTIAL |
| Live Defaulter Dashboard | atriumEngine | getDefaulters, flagOverdueCharges | ServiceChargeMonitor | LIVE |
| Audit Logs & SSO | settings | getFirmActivity | AuditLogViewer (partial) | PARTIAL |
| Dedicated Account Manager | NONE | NONE | NONE | GHOST |
| Custom SLA | NONE | NONE | NONE | GHOST |

Table 3: Pricing Tier Feature Claims vs. Reality

# 2. Ghost and Partial Feature Details

## 2.1 Ghost Features (Marketed but Non-Existent)

### Legacy Intelligence Activation (VEGA)

The landing page dedicates an entire full-width section with animated visuals to this feature, describing physical document digitization, OCR pipelines, and R.A.G. (Retrieval-Augmented Generation) against digitized archives. In reality, there is no code anywhere in the codebase for document digitization. The brainService has a search method that indexes digital content, not physical archives. This is the most elaborate marketing section with zero implementation, representing a serious credibility risk when enterprise clients evaluate the platform during procurement.

### Tenant Vetting System (ATRIUM Pro Tier)

Listed as an "Exclusive" Pro-tier feature at 420,000 Naira per year, the "Tenant Vetting System" consists solely of a manually entered 0-100 score field on the VacancyPipeline lead cards. There is no background check integration, no identity verification, no employment or income validation, no reference checking, and no automated scoring algorithm. The PropertyManagementAgent.ts AI prompt mentions tenant vetting as a capability, but this only influences ALOA chat responses and does not power an actual vetting system. A user paying for this exclusive feature receives only a number input field. Under Nigerian consumer protection law, selling a feature that does not exist could constitute misrepresentation.

### Dedicated Account Manager & Custom SLA (Enterprise)

These are purely contractual and service-oriented features with no software implementation. There is no CRM, assignment system, or even a contact field for an account manager in the schema. While these are service-level commitments rather than software features, listing them alongside functional features on the pricing page creates the impression of a complete product. An enterprise customer evaluating the platform will discover this immediately in a security review, potentially destroying trust in the entire platform.

## 2.2 Critical Partial Features

### ALOA search_legal_repo Returns Empty

The ALOA AI Copilot has a search_legal_repo tool that is stubbed to return an empty array. The backend legal repository query referenced in the code (api.legalRepo.searchLegalRepo) does not exist as a Convex function. When a lawyer asks ALOA to research case law, the tool silently returns zero results. The landing page claims "40+ years of Supreme Court judgments" and badges this as "Live," but the system has procedural rules (timeframes, court jurisdictions) rather than a judgment database. A lawyer missing a precedent because they trusted this claim could face malpractice exposure, and PracticePro could face liability for the misleading "Live" badge.

### SSO Sold as Enterprise Feature but Zero Implementation

The pricing page bundles "Audit Logs & SSO" as a single Enterprise feature. Audit Logs have a partial implementation (a viewer exists behind a paywall), but SSO has absolutely no code. There is no SAML, no OIDC, and no identity provider configuration anywhere in the codebase. An enterprise customer evaluating the platform will discover this immediately during a security review. The bundled promise implies both work, which constitutes a false claim.

### "Automated" Rent Collection is Manual

The landing page claims "Automated Rent Collection" including the ability to "track payments, generate invoices, and reconcile accounts automatically." There is no payment gateway integration (no Paystack, Flutterwave, or bank API). Rent is not collected automatically; it is manually recorded. Invoice generation exists but is basic. Account reconciliation is a manual status change (pending to cleared to defaulted). The word "Automated" is misleading for a product targeting property managers who expect actual payment automation.

### WhatsApp Automation Engine is Simulated

The Pro-tier promises a "WhatsApp Automation Engine" with "Unlimited WhatsApp Reminders." The runDailyAutomation cron job in sentry.ts is simulated. The "How Automations Work" section in the UI describes trigger rules (T-7 reminder, T+1 late notice) that do not actually execute. A property manager expecting automatic rent reminders will find they must manually trigger bulk sends. The feature should be renamed to "WhatsApp Notification Center" or "WhatsApp Composer" until actual event-driven automation exists.

# 3. End-to-End Functional Testing

This section traces the full lifecycle of a Matter (VEGA/Legal) and a Property (ATRIUM/Property) through the DataProvider and domain hooks to verify data persistence across all mutations. Each lifecycle stage is examined for race conditions, rollback correctness, and data integrity.

## 4. State Integrity: Matter Lifecycle

### Create: Race Condition on Client Co-creation

When a new client is created alongside a matter, onAddMatter calls addItem for contacts then addItem for matters sequentially. If the contact addItem succeeds but the matter addItem fails, the contact is already persisted to Convex with no rollback. The error handler in addItem only rolls back the specific table item, not related side effects. Additionally, after matter creation, handleUpdateProperty is called to link the property but is NOT awaited in a try/catch. If the property link update fails, the matter is created but the property is not linked, resulting in silent data inconsistency.

### Read: Phase B Overwrite Risk

The two-phase hydration pattern is well-conceived: Phase A (metadata) loads core list fields in under 500ms, and Phase B (full data) merges the remaining tables. However, Phase B does a blanket state merge that replaces entire table arrays with server data. Any items created optimistically between Phase A and Phase B arrival (typically a 1-3 second window) are silently lost from state. During the initial app load, any matter or property created in the first few seconds after login will disappear from the UI when Phase B data arrives, even though it was successfully persisted to Convex. The data only reappears on the next full query refresh.

### Update: Stale Closure in Optimistic Rollback

The updateItem function captures const prevState = appState before the optimistic update. However, appState is a React state variable in a closure. If two rapid updates occur (e.g., dragging a matter across stages, or bulk property updates), the second update prevState reflects the FIRST update optimistic state, not the true pre-mutation state. On mutation failure, rollback restores the wrong state, potentially corrupting the entire appState for that table. Additionally, there is no version conflict detection; two users editing the same matter simultaneously will silently overwrite each other with last-write-wins semantics.

### Delete: Cascade Delete Never Called from UI

The handleDeleteMatter hook calls actions.deleteItem which only deletes the matter record itself. A dedicated deleteMatterCascade Convex mutation exists that would also remove associated tasks, documents, notes, time entries, expenses, and invoices, but this mutation is NEVER invoked from the UI. The result is that deleting a matter leaves orphaned child records that permanently consume storage and may leak into other views that query by firmId. The archive operation is also non-atomic: if the archive addItem succeeds but the deleteItem fails, the matter exists in both the archive and the matters table simultaneously.

## 5. State Integrity: Property Lifecycle

### CRITICAL: Rent Payments Not Written to ledger_entries

This is the single most dangerous data integrity issue in the application. The CollectRentModal writes rent payments to property.rentPaymentHistory (an embedded array on the property record) and to the invoices table, but does NOT create a ledger_entries record. The getCashFlowSummary query in sentry.ts, which powers the dashboard revenue metrics and cash flow analytics, reads exclusively from ledger_entries. This means ALL rent collection revenue is invisible in financial reports. During high-volume rent season (quarterly collections), potentially millions of Naira in collected rent are missing from the firm revenue dashboard and cash flow analytics. These two data sources can also diverge, showing different totals for the same period.

### Receipt Broken for Rent Payments

The CollectRentModal generates receipts with client: { id: "tenant", name: tenantName }, using a hardcoded "tenant" string as the client ID rather than the actual tenant contact ID. When ReceiptDetailView tries to look up

the client by this ID, it fails and shows an error alert instead of the receipt. Additionally, receipt numbers are generated with REC-${Math.floor(100000 + Math.random() * 900000)}, which is non-deterministic and can produce duplicate numbers under high volume. When no linked matter exists, a dummy matter with id "firm-general" is used for invoice creation, creating reference integrity issues.

### Caution Deposit Dual-Write Inconsistency

When saving a property with a caution deposit, the deposit is routed to ledger_entries via addLedgerEntry in sentry.ts. If this secondary mutation fails, it is caught with console.warn but the property is still saved. The result is that the deposit is recorded in the property but NOT in the ledger. Multi-unit creation is also non-transactional: units are persisted in a for loop with await, and if unit 3 of 5 fails, units 1-2 are persisted while units 3-5 are lost with no rollback.

# 6. Edge Case Analysis

## 6.1 Empty Catch Blocks

| File | Function | Issue | Severity |
|---|---|---|---|
| convex/myFunctions.ts | deleteAccount | Firm deletion failure silently ignored | CRITICAL |
| convex/myFunctions.ts | deleteFirm | Firm record deletion failure silently ignored | CRITICAL |
| convex/myFunctions.ts | deleteMatterCascade | Property lookup failure silently ignored | HIGH |
| src/components/forms/SaveToNoteForm.tsx | Note saving | Error in note saving silently swallowed | HIGH |
| convex/myFunctions.ts | getJoinedFirms | Error when fetching joined firms silently swallowed | MEDIUM |

Table 4: Empty Catch Blocks That Could Cause Silent Failures

## 6.2 Sensitive Console.log Statements

| File | Issue | Severity |
|---|---|---|
| src/contexts/AuthContext.tsx | Leaks firm ID on repair success | HIGH |
| src/contexts/UIContext.tsx | Leaks user IDs on user change | HIGH |
| src/hooks/useBrainAutoIndex.ts | Leaks firm ID during background indexing | HIGH |
| convex/myFunctions.ts | Leaks email address on Brevo send | HIGH |
| src/services/indexer/GeminiStructurer.ts | Leaks AI processing data | MEDIUM |
| convex/myFunctions.ts | Logs firm data loading in Convex | MEDIUM |

Table 5: Console.log Statements Leaking Sensitive Data

### 6.3 Hardcoded IDs and Test Data

| File | Hardcoded Value | Impact | Severity |
|---|---|---|---|
| CollectRentModal.tsx | { id: "firm-general" } | Dummy matter ID for invoices | CRITICAL |
| CollectRentModal.tsx | client: { id: "tenant" } | Breaks ReceiptDetailView | CRITICAL |
| AuthContext.tsx | "atrium-demo-firm-id" | Demo IDs could collide with real data | MEDIUM |
| myFunctions.ts | "demo_firm_id" | Hardcoded demo check in production query | MEDIUM |

Table 6: Hardcoded IDs That Must Be Removed Before Launch

## 7. Performance Audit

The recent "Mega-Query" refactor introduced a two-phase hydration pattern that significantly improves initial load time. Phase A (metadata) unlocks the UI in under 500ms, while Phase B (full data) merges silently. However, several significant performance bottlenecks remain that will cause unacceptable lag as data volume grows.

| Module / Issue | Estimated Impact | Severity |
|---|---|---|
| MatterList O(n2) per-card filtering | 1-2s render with 100 matters | CRITICAL |
| getFirmData 65 DB queries per load | 1.3-2s server time for Phase B | CRITICAL |
| getCashFlowSummary unbounded collect | Timeout with 10K+ entries | CRITICAL |
| MatterList no pagination | 500ms-1s for 100 matters, DOM bloat | CRITICAL |
| Phase B atomic state merge | 200ms JS block freezing main thread | MEDIUM |
| contextActions not memoized | Re-renders all DataActionsContext consumers | MEDIUM |
| 12 nested providers cascade re-renders | Cascading re-renders on any mutation | HIGH |
| Dashboard unmemoized stats | 100ms per re-render with 50 properties | MEDIUM |
| RevenueEngine inline ledger filters | 50ms per render with 1000 entries | MEDIUM |
| PropertyManagerView no pagination | DOM bloat with 50+ properties | MEDIUM |
| handleBulkUpdateProperties N parallel mutations | 50 HTTP requests for 50 property edits | MEDIUM |
| ReportingView synchronous report generation | 500ms-2s UI freeze during generation | MEDIUM |

Table 7: Performance Bottleneck Summary

### Estimated Total Load Time at Scale

With 100 matters, 50 properties (500 units), and 1,000 ledger entries, the estimated total time to fully interactive is 3-5 seconds. Phase B (1.5-2.5s server time) and MatterList rendering (500ms-1.5s due to O(n2) per-card filtering) are the worst offenders. The Phase A metadata projection is well-optimized at 400-600ms. The primary fix should target the MatterList per-card filtering pattern: pre-building Map of matterId to Task/Document arrays would

reduce render from 1.5s to approximately 50ms.

# 8. Nigeria-Specific Feature Gaps

Based on the current codebase analysis and the Nigerian property management market, the following critical "last-mile" features are missing. Each gap is classified by its impact on the Nigerian market and the current implementation status.

| Feature | Nigeria Market Need | Current Status | Priority |
|---|---|---|---|
| WHT on Rent (10%) | Mandatory under PITA for corporate landlords | PARTIAL: VEGA only, not in Atrium | P0 |
| VAT on Service Charges (7.5%) | Required by VAT Act 2007 | PARTIAL: Invoice forms only, not service charges | P0 |
| Quit Notice Automation | Lagos Tenancy Law 2011 requires statutory notices | CRITICAL MISSING | P0 |
| Tenancy Agreement Generation | Every tenancy requires written agreement | CRITICAL MISSING | P0 |
| PDF Receipt Generation | Tenants expect formal receipts for tax documentation | PARTIAL: Bare HTML print only | P1 |
| Auto-Penalty for Late Rent | Late rent is endemic in Nigeria | PARTIAL: Manual 5% click-to-apply only | P1 |
| Land Use Charge Compliance | Mandatory for Lagos State property owners | CRITICAL MISSING | P1 |
| CAM Allocation Engine | Multi-tenant buildings need fair charge splitting | CRITICAL MISSING | P2 |
| Bank Reconciliation | Multiple payment channels need reconciliation | CRITICAL MISSING | P2 |
| Multi-Currency Support | Ikoyi/VI/Maitama properties charge in USD | CRITICAL MISSING | P2 |

Table 8: Nigeria-Specific Feature Gap Analysis

The most critical gap is the absence of Withholding Tax (WHT) and Value Added Tax (VAT) tracking on Atrium property collections. Every single rent payment and service charge in Nigeria has tax implications. WHT at 10% on rent to corporate landlords is mandatory under PITA, and VAT at 7.5% on service charges is required by the VAT Act 2007 as amended in 2020. Current Atrium has zero tax tracking on property collections, which means firms using the platform cannot generate compliant tax reports for FIRS filing. The Quit Notice Automation gap is equally dangerous: missing a quit notice deadline means you cannot legally recover possession, and the case will be dismissed in court. The AI agent already knows the law; it just needs execution capability.

# 9. Technical Debt Report

## 9.1 Code Quality Regression

### Dual Payment Systems

Properties track rentPaymentHistory (an array on each property record) while Atrium separately tracks ledger_entries (a dedicated table). No synchronization mechanism exists. A rent payment recorded via CollectRentModal updates rentPaymentHistory but does NOT create a ledgerEntry. These are two independent financial records with no single source of truth, creating a scenario where tenants could appear paid in one system but overdue in another.

### Auth Pattern Duplication

Three separate auth mechanisms exist: (1) withFirmAuth, an HOC wrapper that looks up users by firmId but never calls getUserIdentity(); (2) authUtils.ts with PBKDF2 password hashing for login only; (3) authHelpers.ts with requireFirmUser() that does call getUserIdentity() and is the correct pattern. Worse, withFirmAuth is not used by any mutation (zero references in myFunctions.ts) and withFirmAuth allows a "skip" string to bypass auth entirely. The withFirmAuth function gives a false sense of security because it does not verify WHO is calling, only that some user exists with the given firmId.

### Type Safety: The "any" Epidemic

The codebase contains approximately 680 instances of "any" type usage across 90+ files. The Convex schema has 86 instances of v.any() validation bypass, with 66 in schema.ts alone. The three generic CRUD mutations (createItem, updateItem, deleteItem) accept data: v.any(), meaning any data shape can be written to any table with zero validation. Key fields like rentalDetails, disputeDetails, saleDetails, specialtyData, parties, bankAccounts, rentPaymentHistory, and metadata are all v.any(). This compromises database integrity and eliminates any migration path for schema changes since there is no schema to evolve.

### Context State Overlap

Five domain contexts (CoreContext, MatterContext, FinanceContext, DocumentContext, ExecutionContext) are pure passthroughs that each destructure appState from DataContext, re-slice it, and re-expose the same updateItem/deleteItem from DataContext. This adds 5 provider layers with zero added logic. The DataContext itself has an ExtendedDataActions interface with 70+ methods, composing 8 domain hooks into one giant contextActions object. Any change to any hook triggers re-evaluation of the entire context value.

# 10. Pre-Launch Blockers

## 10.1 Hardcoded API Keys in Client Bundle

Three unique Gemini API keys are hardcoded in client-side code. The key AIzaSyAjPumBNTGi8Lzg457yKm4dD0jFAzefXo0 appears in aiUtils.ts (line 163), AloaChat.tsx (line 464), and geminiService.ts (lines 246, 379, 460, 544). A second key AIzaSyDd5ib2A1562gO2PY1FQElSVzwyIaeBAN8 also appears in geminiService.ts. Additionally, the embedding API URL in aiUtils.ts appends the key as a query parameter (?key=${apiKey}), which means API keys are logged by proxies, CDNs, and browsers. All AI calls must be routed through Convex actions (server-side) where keys remain secret.

## 10.2 Zero Authentication on CRUD Mutations

The three most critical mutations, createItem, updateItem, and deleteItem, have zero authentication. No getUserIdentity(), no withFirmAuth, no requireFirmUser. Any client can call deleteItem({ table: "users", id: "..." }) and delete any record. This is the single highest-risk security issue in the application and must be fixed before any

paying user logs in.

### 10.3 Offline Admin Bypass

When Convex is unreachable, the app creates a mock offlineUser with role: "Admin", firmId: "offline_firm", and isVerified: true. This means anyone can get admin access by simply disconnecting from the network. This must be replaced with cached JWT verification or limited to read-only mode.

### 10.4 Demo/Mock Code Not Gated

The demo@practicepro.ng bypass, VEGA_DEMO_APP_STATE, ATRIUM_DEMO_APP_STATE, and FloatingTestControls are present in production code. The demo user gets free AI usage, bypassed auth, and special UI treatment. The isDemo check is scattered across 13+ files. All demo code must be gated behind import.meta.env.DEV or a backend-controlled feature flag to ensure it is excluded from production builds.

# 11. Launch Readiness Checklist

The following checklist consolidates all findings into three priority categories. Items marked [Must Fix] will cause data loss, security breaches, or legal liability if not addressed before the first paying user. Items marked [Remove for V1] should be hidden or stripped from the product to avoid misleading customers. Items marked [Optional for V1] are recommended improvements that enhance the product but do not block launch.

## MUST FIX Before V1

| Item | Category | Est. Effort |
|---|---|---|
| Wire requireFirmUser() into createItem/updateItem/deleteItem mutations | Security | 8h |
| Remove all 3 hardcoded Gemini API keys; route AI calls through Convex actions | Security | 6h |
| Kill offline Admin bypass in AuthContext.tsx; replace with cached JWT or read-only mode | Security | 4h |
| Remove "skip" auth bypass string from withAuth.ts | Security | 1h |
| Gate all demo/mock code behind import.meta.env.DEV | Security | 8h |
| Fix rent collection: write to ledger_entries alongside rentPaymentHistory | Data Integrity | 4h |
| Fix ReceiptDetailView for rent receipts (client.id = "tenant" breaks lookup) | Data Integrity | 3h |
| Wire deleteMatterCascade from UI instead of simple deleteItem | Data Integrity | 2h |
| Fix Phase B hydration overwrite of optimistic creates | Data Integrity | 4h |
| Fix stale closure in updateItem rollback (use ref-based snapshot) | Data Integrity | 3h |
| Change "Tenant Vetting System" from Exclusive Pro-tier claim (it is just a manual score input) | Legal/Marketing | 1h |
| Remove "40+ years of Supreme Court judgments" claim; change "Live" badge to "Beta" | Legal/Marketing | 1h |
| Split "Audit Logs & SSO" into separate line items; mark SSO as "Coming Soon" | Legal/Marketing | 1h |
| Remove "Legacy Intelligence Activation" section from landing page | Legal/Marketing | 1h |

| | | |
|---|---|---|
| Rename "WhatsApp Automation Engine" to "WhatsApp Notification Center" | Legal/Marketing | 1h |
| Remove "Automated" from "Automated Rent Collection" | Legal/Marketing | 1h |
| Add per-route ErrorBoundary wrapping for all detail views | Reliability | 6h |
| Fix empty catch blocks in deleteAccount, deleteFirm, deleteMatterCascade | Reliability | 2h |
| Remove sensitive console.log statements (firm IDs, user IDs, emails) | Security | 2h |
| Remove FloatingTestControls from production build | Security | 1h |

Table 9: Must Fix Before V1 (20 items)

## REMOVE For V1

| Item to Remove or Relabel | Reason |
|---|---|
| Legacy Intelligence Activation landing page section (entire block) | Ghost Feature |
| "Dedicated Account Manager" from Enterprise pricing cards | Ghost Feature |
| "Custom SLA Guarantee" from Enterprise pricing cards | Ghost Feature |
| "Tenant Vetting System" as Pro-tier exclusive (or rename to "Applicant Pipeline") | Ghost Feature |
| ALOA search_legal_repo tool (returns empty results silently) | Broken Feature |
| "Bank-grade encryption" and "AES-256 at Rest" claims (add asterisk: provided by Convex) | Misleading |
| Trust badge "ISO 27001 Aligned" without supporting documentation | Misleading |
| VacancyPipeline vetting score field (replace with proper vetting workflow or remove) | Incomplete |

Table 10: Remove or Relabel for V1 (8 items)

## OPTIONAL For V1

| Item | Category | Est. Effort |
|---|---|---|
| MatterList O(n2) performance fix (pre-build Map, pagination) | Performance | 1w |
| Dashboard and RevenueEngine useMemo wrapping | Performance | 2d |
| Replace v.any() on critical schema fields with typed validators | Type Safety | 2w |
| Consolidate 5 passthrough contexts into direct DataContext access | Architecture | 1w |
| Extract shared form abstraction (ALOA listener, firmId resolution) | Code Quality | 3d |
| Add WHT/VAT fields to Atrium ledger_entries and service_charges | Nigeria Compliance | 1.5w |
| Add quit notice automation with Lagos Tenancy Law 2011 compliance | Nigeria Compliance | 2w |
| Build tenancy agreement document generation | Nigeria Compliance | 3w |
| Proper PDF receipt generation with firm branding and tax breakdown | User Experience | 1.5w |

| | | |
|---|---|---|
| Auto-penalty logic for late rent payments | Nigeria Compliance | 1w |
| Land Use Charge tracking for Lagos properties | Nigeria Compliance | 1.5w |
| CAM allocation engine for multi-tenant buildings | Nigeria Compliance | 2w |
| Bank statement reconciliation engine | Nigeria Compliance | 3w |
| Multi-currency support (USD rent) | Nigeria Compliance | 2w |

Table 11: Optional for V1 (14 items)

## Audit Summary Statistics

| Metric | Count |
|---|---|
| Ghost Features (marketed, no code) | 4 |
| Partial Features (UI exists, incomplete) | 13 |
| Live Features (fully functional) | 7 |
| Critical Security Vulnerabilities | 7 |
| Hardcoded API Keys in Client Bundle | 3 (appearing ~10 times) |
| Empty Catch Blocks (silent failures) | 7 |
| Console.log Leaking Sensitive Data | 9 |
| Hardcoded IDs in Production Code | 4 |
| Total "any" Type Usages | ~680 across 90+ files |
| Total v.any() in Convex Schema | 86 (66 in schema.ts) |
| Mutations Without Authentication | 3 critical (createItem, updateItem, deleteItem) |
| Context Providers in React Tree | 12 |
| Must Fix Items | 20 |
| Remove for V1 Items | 8 |
| Optional for V1 Items | 14 |

Table 12: Audit Summary Statistics